

Muhammad Hamza Tahir

📍 Lahore, Punjab, Pakistan ✉ m.hamza.codes@gmail.com ☎ +923147838060 🌐 in/m-hamza-tahir-

SUMMARY

Versatile **Software Engineer** with experience across **Embedded Systems, IoT Firmware (ESP32), and Python/Django Backend Development**. Skilled in designing secure communication systems, building full-stack applications, developing scalable server-side architectures, and optimizing real-time IoT pipelines. **Open to global remote opportunities.**

EXPERIENCE

Development Engineer

Robusys

September 2025 – Present, Lahore, Pakistan

- **Developed** firmware for ESP32 and microcontroller-based systems, improving device reliability by **30–40%** through optimized communication and memory handling.
- **Drafted** patent documentation and technical descriptions, contributing to **3–5 early-stage IP filings** and structured innovation workflows.
- **Conducted** market and competitive research across **20+ sources**, supporting product and R&D decisions with actionable insights.
- **Authored** technical documentation, process flows, and structured reports, reducing engineering review time by **25%**.

Technical Research & Development Associate

Logithoroughfare AB

January 2024 – August 2025, Sweden

- **Conducted** detailed market, technical, and competitor research across **20+ sources**, providing insights that supported product and R&D decision-making.
- **Created** and improved web content, landing pages, and internal tools, increasing content clarity and usability by **30%** based on internal feedback.
- **Drafted** structured technical documents and **3–5 initial patent drafts**, enhancing the company's IP documentation pipeline.
- **Produced** clear research summaries and documentation packs, reducing review time for engineering teams by **25%**.
- **Organized** research findings into actionable insights that helped guide product roadmaps and technical planning.
- **Implemented** technical diagrams, workflows, and system documentation to improve internal communication and project clarity.

Backend Intern

Enigmatix

November 2023 – December 2024, Bahawalpur

Enigmatix is a software development company that provides SaaS, ERP, and AI solutions.

- **Completed** a backend internship focused on Python, Django, and full-stack development, contributing to **3+ production-level modules** and improving code quality through peer reviews.
- **Designed and developed 2–3 full-stack web projects**, implementing complete front-end (HTML, CSS, Bootstrap, JS) and backend features, achieving **100% functional requirements coverage** in project evaluations.
- **Utilized** Django ORM for efficient database interaction, **reducing query complexity by ~25%**, while implementing validation workflows that decreased user input errors by **40%**.

PROJECT

ExplorOut – Full-Stack E-Commerce

Freelancing • [explorout.com](#) • June 2025 – September 2025

- Developed a fully functional e-commerce platform with authentication, product management, cart, checkout, and order workflows.
- Designed responsive UI and reusable templates, improving user experience and layout consistency by **30%**.
- Deployed on a Hosting Platform, achieving **99% uptime** during testing.

Smart Solar Inverter – Real-Time IoT Monitoring System

FYP – The Islamia University of Bahawalpur • December 2024 – June 2025

- Engineered end-to-end IoT system for AC/DC power monitoring with **<2-second** dashboard latency.
- Designed firmware for stable sensor readings and optimized communication, reducing packet loss by **40%**.
- Created real-time Django dashboard handling **2,000+ live datapoints**, improving monitoring accuracy for field testing.

Vaccine Scheduling & Booking System

Enigmatix • August 2024 – November 2024

- Developed role-based appointment system (Admin/Patient) with conflict detection.
- Implemented smart slot management, reducing booking overlaps by **~80%**.
- Designed clean UI and session-handling workflow, improving user experience and error reduction.

SKILLS

Technical Skills: Python, Django, RestAPI, C, Embedded Programming, Firmware Development, Microcontrollers, ESP32, FreeRTOS, JavaScript, JQuery, AJAX, WebSockets, Real-time systems, Git, Basic CI/CD

Soft Skills: Problem Solving, Analytical Thinking, Adaptability & Multitasking, Technical Communication, Remote Collaboration, Attention To Detail, Structured Documentation, Self-Learning

Languages: English – Professional Working Proficiency, Technical English, Urdu – Native

EDUCATION

Bachelor's in Computer Science

The Islamia University of Bahawalpur • Bahawalpur • 2025 • 3.32
